Spectral-Local Residual Networks for Wave Inverse Reconstruction

May 7, 2026

Abstract

Wave inverse problems like coherent diffractive imaging (CDI) require networks to combine global measurement-domain coupling with sharp real-space synthesis. Pure-convolutional and pure-spectral models each resolve this tension only partially. We show that closing this gap on simulated CDI requires getting two choices right at once: training through the known diffraction model, and placing spectral mixing in the encoder rather than the bottleneck of an otherwise local architecture. Switching from object-space supervision to a PtychoPINN-style measurement-consistency loss reduces factorized Fourier neural operator (FFNO) amplitude mean absolute error (MAE) more than fourfold ($0.35 \rightarrow 0.08$), with consistent gains across every architecture we train both ways. Holding that loss fixed, we introduce **SRU-Net** (spectral-residual U-Net), which interleaves truncated Fourier mixing with local convolution in each encoder block, processes the result through a six-block residual bottleneck, and decodes with transposed convolutions. SRU-Net cuts CDI amplitude MAE 3.0× over FFNO and 4.6× over a convolutional neural network (CNN) baseline (0.027 vs. 0.082 vs. 0.123) and reaches amplitude structural similarity index (SSIM) 0.99. Ablations indicate the gain depends on where spectral mixing is placed, not on spectral capacity alone: replacing the residual bottleneck with FFNO-style spectral blocks or weakening the decoder regresses both error metrics. The same design transfers to Born–Rytov diffraction tomography (BRDT), where SRU-Net reaches relative L_2 of 0.288 against 0.451 for FFNO and 0.380 for a model-based Born inverse. On forward compressible Navier–Stokes prediction, FFNO remains strongest; SRU-Net is second by aggregate error, while U-shaped neural operator (U-NO) is close to Fourier neural operator (FNO) in aggregate error and stronger at higher frequencies.

1 Introduction

Problems such as phase retrieval [6], inverse scattering [8], and forward partial differential equation (PDE) prediction combine long-range coupling with localized structure. In lensless computed imaging, each detector measurement depends on the target object and probe interaction through global wave propagation, while the reconstruction is a real-space object with localized amplitude and phase structure. In fluid dynamics, large-scale coherent dynamics coexist with localized features such as shocks and vortices. Across these settings, learned models must propagate information globally without erasing localized spatial structure.

Classical forward solvers often require fine discretization, while inverse wave reconstruction commonly relies on iterative physics-based reconstruction or projection algorithms. Standard computer vision architectures adapted to physics-prediction tasks provide fast approximate solutions, but their accuracy is limited in domains with prominent scale mixing. The accuracy gap reflects a specific architectural mismatch: architectures that are efficient on natural images, such as U-Net [18] and ResNet [17], process local spatial neighborhoods well but lack direct global receptive fields for long-range physical interactions.

Fourier neural operator (FNO) networks address the opposite side of this tradeoff. They are effective models for continuous physical systems because spectral layers provide global mixing [10]. However, FNO-style models retain a finite set of Fourier modes, and this truncation can bias predictions toward lower-frequency structure and weaken sharp localized features. Recent extensions mitigate this limitation by coupling global Fourier layers with U-Net-style [18] or localized-kernel branches [12, 11]. These convolutional or localized-kernel paths help preserve fine-scale structure while retaining long-range mixing. Such hybrids improve some PDE benchmarks, but they are not uniformly better than strong spectral-only architectures such as the factorized Fourier neural operator [13]. We focus on convolutional and spectral models; attention-based operators such as Galerkin Transformer [16] and the broader DeepONet family [15] are separate model families outside the scope of this study.

Coherent diffractive imaging (CDI) is our primary test case because it stresses both sides of the global-local problem. Diffraction measurements are globally coupled by the wave forward model, while the desired output is a localized complex object. The inverse map must therefore interpret measurement-domain structure across the full detector field and synthesize a sharp real-space reconstruction. Our experiments show that FNO-based architectures used in forward physical prediction can struggle to synthesize accurate real-space images from diffraction measurements, although they still improve over purely convolutional models in several comparisons.

For CDI, the training objective is as important as the architecture. We do not propose a new CDI loss; instead, we use the established PtychoPINN-style physics-consistency objective, in which a predicted object is passed through known real-space consistency and diffraction operators and the loss is applied in measurement space [1]. This objective is not a minor implementation detail in our experiments. Models trained with supervised loss show meaningfully worse aggregate reconstruction quality than the corresponding physics-consistency-trained models across the tested architectures, with the clearest gaps in amplitude reconstruction. We therefore use the same physics-consistency objective for the CDI architecture comparison and ask which network works best once the standard CDI physics constraint is imposed.

We introduce SRU-Net, a spectral-convolutional architecture for wave inverse reconstruction. Each encoder block combines a truncated Fourier mixing branch with a local convolutional branch inside a residual update. The Fourier branch lets the model aggregate information across the full measurement field, while the convolutional branch preserves grid-aligned local features. A residual bottleneck processes the mixed representation at reduced spatial resolution, and a convolutional decoder synthesizes the real-space output. The architecture is designed to separate the inverse-mapping role of global spectral mixing from the image-synthesis role of local convolutional decoding.

Our contributions are threefold. *First*, we argue that for wave inverse reconstruction, spectral mixing belongs in the encoder, paired with a local residual bottleneck and a convolutional decoder; we instantiate this as SRU-Net. *Second*, on CDI under a shared PtychoPINN-style physics-consistency objective, SRU-Net cuts amplitude MAE to 0.027, a 3.0× improvement over FFNO (0.082) and 4.6× over a CNN baseline (0.123), and reaches amplitude SSIM 0.99. The same objective reduces amplitude error for every architecture trained both ways, isolating the architectural gap from the loss. *Third*, ablations isolate the placement: the convolutional decoder is load-bearing (replacing it with a linear decoder drives amplitude error to 5× the SRU-Net value), and the local residual bottleneck cannot be replaced by spectral blocks without regressing both amplitude and phase metrics.

2 Related Work

2.1 Operator learning and spectral-local architectures

Neural operators learn mappings between function spaces and are widely used for scientific machine learning on PDEs. Fourier neural operators (FNOs) parameterize solution operators through truncated spectral convolutions, providing global receptive fields at low cost [10]. Factorized Fourier neural operators (FFNOs) extend this construction with separable Fourier transforms, layer-shared kernel integral operators, and residual temporal updates, enabling deeper and more parameter-efficient stacks [13]. PDEBench supplies a controlled benchmark for evaluating these models on forward physical prediction, including the compressible Navier–Stokes setting we use here [14].

Pure spectral models couple distant regions of a field directly, which is attractive for wave propagation and fluid dynamics. The price is Fourier-mode truncation, which biases predictions toward lower-frequency structure and can attenuate sharp localized features. Several extensions address this by combining global Fourier layers with local or multiscale operators: U-FNO interleaves spectral and U-Net branches for multiphase flow [12], and neural operators with localized integral and differential kernels pair spectral coupling with locally supported kernels [11]. These hybrids improve specific PDE benchmarks but do not uniformly outperform strong spectral-only baselines such as FFNO, and the literature has not converged on where in an architecture spectral and local components should be placed.

2.2 Wave inverse reconstruction

Coherent diffractive imaging and ptychography have a long algorithmic history grounded in iterative projection methods: the Gerchberg–Saxton algorithm, hybrid input–output (HIO), error reduction (ER), the difference-map algorithm [6], and ePIE [7]. These methods are widely used in practice but can require many iterations to converge, and each reconstruction is solved independently. Learned reconstruction replaces per-instance iterative solving with a one-shot inverse map at inference time, motivating recent CDI work in machine learning that ranges from supervised generative inversion [4, 2], through edge-streaming pipelines [5], to unsupervised physics-aware variants that embed the forward model in the loss [3, 1].

PtychoPINN introduced a measurement-consistency training objective for unsupervised scanning-CDI [1]: a learned network predicts complex object patches; the prediction is passed through known real-space consistency, probe interaction, and far-field diffraction; and the loss compares the resulting simulated detector amplitudes to the measurements. Training is therefore against the measured diffraction statistics rather than against paired target objects. We adopt this formulation unchanged for the CDI portion of our study.

The reason measurement-space objectives are natural for CDI is structural and predates deep learning. In the variational picture of inverse problems, the standard objective combines a measurement-space data-fidelity term with an image-space regularizer: the data term is invariant to changes in the prediction that the forward operator does not produce in the measurement, while the regularizer carries prior information about the reconstruction. A supervised object-space loss collapses these two roles into a single image-space penalty, which can penalize nuisance variation unless the supervised target or loss handles those equivalences explicitly. Unrolled iterative methods such as ISTA-Net and ADMM-Net build this separation into the architecture by interleaving measurement-domain and image-domain updates; we use the same intuition implicitly through measurement-space training rather than through architectural unrolling.

2.3 Forward versus inverse problems in scientific ML

Operator-learning benchmarks are dominated by forward physical prediction: given current state and parameters, predict the next state on the same grid. Wave inverse reconstruction differs in that the input occupies measurement space and the output occupies image space, coupled by a global forward operator. Architectures optimal for forward prediction need not be optimal for inverse mapping, and the converse holds as well. Our PDEBench Navier–Stokes results are reported in this spirit: as context that constrains where SRU-Net's design helps, not as a competing claim on forward prediction.

3 Methods

3.1 Task-agnostic network formulation

Let x denote the task input and y the target field or reconstruction. We write the learned network as

$$\hat{y} = G_\theta(x). \tag{1}$$

For CDI, x is a stack of diffraction measurements and $\hat{y}$ is a complex object-patch representation, meaning local crops of the reconstructed complex object represented by real and imaginary channels. For Born–Rytov diffraction tomography (BRDT), x is a Born image formed by backprojecting the measured complex sinogram, and $\hat{y}$ is the recovered scattering potential. For PDEBench compressible Navier–Stokes (CNS), x is a temporal history of physical fields and $\hat{y}$ is the predicted next state. The tasks use different input representations and objective functions, but share the same architectural principle: G_θ combines global spectral mixing with local convolutional feature recovery.

We decompose the network as

$$G_\theta = D_\theta \circ B_\theta \circ E_\theta, \tag{2}$$

where E_θ is the spectral-local encoder, B_θ is a ResNet bottleneck at reduced spatial resolution, and D_θ is a convolutional upsampling decoder.

We use subscripts to distinguish architectural positions rather than tensor dimensions. The index ℓ denotes the encoder block that maps z_ℓ to $z_{\ell+1}$; j indexes the six residual bottleneck blocks; s indexes resolution levels used for encoder-decoder skip connections; and k indexes Fourier modes. The retained Fourier-mode set is K, with $m = |K|$ in the architecture diagram. Superscripts such as $B_j^{\times 6}$ denote repeated blocks rather than exponentiation. Task subscripts such as CDI and CNS label task-specific outputs.

3.2 Spectral-local encoding

For a feature tensor z_ℓ, a spectral-local encoder block has the form

$$z_{\ell+1} = \sigma\left(L_\ell z_\ell + \mathcal{F}^{-1}\left[R_\ell(k)\,\mathcal{T}_K\left(\mathcal{F}(z_\ell)\right)(k)\right]\right), \tag{3}$$

where L_ℓ is a learned local convolutional map, $\mathcal{F}$ is the 2D Fourier transform, $\mathcal{T}_K$ retains a fixed set of Fourier modes, $R_\ell(k)$ is a learned complex spectral mixing tensor over retained modes, and σ is the Gaussian error linear unit (GELU) nonlinearity [19] used in the implemented SRU-Net. In residual form, this can be written as

$$z_{\ell+1} = z_\ell + \sigma\left(L_\ell z_\ell + S_\ell z_\ell\right), \tag{4}$$

with S_ℓ denoting the truncated spectral operator. The local path preserves high-frequency and edge-like information that may be weakened by spectral truncation, while the spectral path provides global coupling.

3.3 Spectral-convolutional residual architecture

The architecture separates three roles. Spectral mixing in the encoder couples information across the full measurement field, where global aggregation is most useful for the inverse map. The bottleneck processes the resulting representation with local residual blocks at reduced spatial resolution. The convolutional decoder then synthesizes the real-space output. This separation reflects the hypothesis that global mixing is most useful during measurement-to-object mapping, while local convolutional processing is most useful for object-space refinement and output synthesis.

The implemented SRU-Net consists of an initial convolution, four spectral-local encoder blocks, two downsampling stages, a six-block residual bottleneck at $N/4$ spatial resolution, and a convolutional upsampling decoder. Figure 1 summarizes the data flow, and Table 1 lists the architectural choices that determine the shape of the model.

In the architecture diagram, ϕ_0 is the initial channel projection, E_0 through E_3 are encoder blocks, B_j denotes a residual bottleneck block, and U_1, U_2 are decoder stages. The spatial resolution is denoted by N, C_{in} is the task-specific input channel count, m is the number of retained Fourier modes, and e_s, d_s are encoder and decoder features at scale s. In the CDI configuration, each decoder stage applies a transposed convolution with kernel size 4, stride 2, and padding 1, followed by instance normalization and GELU. When skip-add ablations are enabled, a learned 1×1 convolution maps encoder features into the decoder channel space before additive fusion. Within encoder block E_ℓ, z_ℓ is the input feature tensor, $S_\ell = \mathcal{F}^{-1} R_\ell \mathcal{T}_K \mathcal{F}$ is the truncated spectral operator, L_ℓ is the local convolutional path, and σ is GELU. Here $\mathcal{F}$ is the Fourier transform, $\mathcal{T}_K$ keeps the retained modes, and R_ℓ mixes those modes. Within bottleneck block B_j, u_j is the bottleneck feature tensor, $R_j = \text{Conv}_{3\times3} \cdot \text{InstNorm} \cdot \text{GELU} \cdot \text{Conv}_{3\times3} \cdot \text{InstNorm}$ is applied before the residual addition, and α_j is its learned residual scale; the centered dot lists operations in left-to-right order. Here $\text{Conv}_{3\times3}$ denotes a 3-by-3 convolution. Within CDI decoder stage D_s, the coarser decoder state is updated by $\text{ConvT}_{4,s2,p1} \cdot \text{InstNorm} \cdot \text{GELU}$. The optional skip-add path first applies a 1×1 convolution to the matching encoder feature before fusion.

When skip connections are enabled, encoder features e_s are fused into decoder states d_s. For additive fusion,

$$d_s = U_s(d_{s+1}) + \text{Conv}_{1\times1}(e_s), \tag{5}$$

where U_s is the decoder-stage upsampling operator and the 1×1 convolution maps the same-scale encoder feature e_s into the decoder channel space. For concatenative fusion,

$$d_s = \text{Conv}_{1\times1}\left([U_s(d_{s+1}), e_s]\right). \tag{6}$$

The CDI network outputs complex object patches in real/imaginary format. In BRDT, the same reconstruction architecture outputs a single real scattering potential. In the PDEBench experiments, the architecture is evaluated on real-valued physical field histories and next-state targets rather than diffraction measurements.

3.4 CDI physics-consistency objective

For CDI, SRU-Net predicts local complex object patches,

$$\hat{o} = G_\theta(x). \tag{7}$$

Table 1: **SRU-Net architecture summary.** The same architecture is used for CDI reconstruction and CNS field prediction, with task-specific input representations, output variables, and objective functions.

Component	Implementation details
Input representation	CDI uses grouped object-patch channels and returns complex real/imaginary patches. CNS uses real-valued field histories as input and predicts the next density, velocity, and pressure fields. In both settings, the first convolution maps the input to width 32.
Encoder	Four spectral-local blocks. Each block adds a learned Fourier-mode mixing branch and a reflected-padding 3×3 convolution branch inside an outer residual update; the branch sum is passed through GELU before the outer residual addition. The retained-mode count m is a configurable parameter; the default setting uses $m = 12$ unless otherwise stated. The encoder downsamples after the first two blocks, giving $N \to N/2 \to N/4$ and widths 32, 64, and 128.
Bottleneck	Six constant-resolution residual blocks at $N/4$ in the He et al. sense [17]. Each block uses a reflected-padding 3×3 convolution, instance normalization [20], GELU [19], a second reflected-padding 3×3 convolution, instance normalization, and a learned residual scale. The spectral-residual variant replaces this local bottleneck with residual blocks that add a shared factorized spectral branch.
Decoder and skips	The CDI path uses two decoder stages in the U-Net encoder-decoder family [18], each $\mathrm{ConvT}_{4,s2,p1}$ followed by instance normalization and GELU. The CNS model uses two pixel-shuffle [21] upsampling stages and additive encoder-decoder skips from the $N/2$ and N encoder resolutions, with each skip feature first adapted by a learned 1×1 convolution.
Output and loss	CDI predicts real/imaginary complex patches and trains through the coherent diffractive imaging consistency and diffraction operators. BRDT predicts a real scattering potential and trains with supervised object loss plus Born consistency. CNS predicts real field channels and trains against the next physical state.

The prediction is not trained only by comparing $\hat{o}$ to a paired object target. Instead, it is passed through a real-space consistency operator F_c, which stitches overlapping local patches into a common object canvas and re-extracts position-consistent patches. The consistent object patches are then passed through a known diffraction map F_d:

$$\hat{x} = F_d\left(F_c(\hat{o}; r); P\right), \tag{8}$$

where r denotes scan positions and P is the probe. The CDI objective is defined in measurement space:

$$\mathcal{L}_{\mathrm{CDI}} = \mathcal{D}_{\mathrm{amp}}\left(x, F_d(F_c(G_\theta(x); r); P)\right). \tag{9}$$

The reported CDI runs use detector-amplitude mean absolute error. The stored detector tensors are diffraction amplitudes, so after applying detector scaling the loss is

$$\mathcal{D}_{\mathrm{amp}}(x, \hat{x}) = \frac{1}{\bar{x} + \epsilon} \frac{1}{|\Omega|} \sum_{i \in \Omega} |\hat{x}_i - x_i|, \tag{10}$$

where x_i and $\hat{x}_i$ are measured and predicted detector amplitudes, $\bar{x}$ is the mean measured detector amplitude in the batch, Ω indexes detector pixels and scan positions, and ϵ prevents numerical singularities. This is the physics-consistency objective used in the CDI architecture comparison and in the supervised-loss comparisons.

3.5 CDI supervised objective

To isolate the role of the training loss, we also train models with a supervised object-space loss when the same architecture was also trained with the CDI forward model. These runs use the same input data, architecture, optimizer, number of epochs, and evaluation metrics as their physics-consistency-trained counterparts, but the loss is applied directly to the predicted object representation rather than after the CDI forward model. The main architecture comparison uses the shared physics-consistency objective.

The supervised objective has the form

$$\mathcal{L}_{\text{sup}} = \mathcal{D}_{\text{obj}} \left(G_\theta(x), y \right), \tag{11}$$

where y is the object target and $\mathcal{D}_{\text{obj}}$ is the object-space reconstruction loss used by the corresponding supervised run. Metrics are computed after the same evaluation alignment and post-processing used for physics-consistency-trained models.

3.6 Born–Rytov diffraction tomography

We also evaluate SRU-Net on a Born–Rytov diffraction tomography (BRDT) inverse-scattering task. The target is the physical scattering potential

$$q(x, z) = k_m^2 \left[\left(\frac{n(x, z)}{n_m} \right)^2 - 1 \right], \tag{12}$$

where $n(x, z)$ is the refractive-index field, n_m is the background medium index, and k_m is the in-medium wavenumber. Under the first Born approximation [8], each incident plane wave produces a scattered field whose far-field samples are linear measurements of the Fourier transform of q on the corresponding Ewald arc. We write this discrete forward map as

$$s = \mathcal{B}(q), \tag{13}$$

where $s \in \mathbb{R}^{A \times D \times 2}$ is the complex sinogram over illumination angles and detector frequencies, represented by real and imaginary channels.

The BRDT learned-model input is the measured complex sinogram itself, represented as real and imaginary channels and reordered to a two-channel detector-angle tensor x_{BRDT}. The model predicts a single-channel scattering potential,

$$\hat{q}_\theta = G_\theta(x_{\text{BRDT}}). \tag{14}$$

All neural models use that same sinogram input representation and the same differentiable Born forward operator during training. The model-based Born inverse remains a separate non-learned reference and is not used as learned model input.

The BRDT objective combines supervised reconstruction with measurement-space Born consistency, plus total variation (TV) [9] and positivity regularization:

$$\begin{aligned} \mathcal{L}_{\text{BRDT}} = \lambda_{\text{img}} \left\| \hat{q}_\theta^{\text{norm}} - q^{\text{norm}} \right\|_1 + \lambda_{\text{abs}} \left\| \mathcal{B}(\hat{q}_\theta) - s \right\|_1 \\ + \lambda_{\text{rel}} \frac{\left\| \mathcal{B}(\hat{q}_\theta) - s \right\|_2}{\left\| s \right\|_2 + \epsilon} + \lambda_{\text{tv}} \, \text{TV}(\hat{q}_\theta) + \lambda_{\text{pos}} \left\| \min(\hat{q}_\theta, 0) \right\|_2^2 . \end{aligned} \tag{15}$$

The image loss is computed in the normalized target space used by the model. Before applying the Born operator, predictions are converted back to physical q units using training-set normalization statistics. This prevents the measurement-consistency term from being evaluated on normalized, dimensionless outputs. We use $\lambda_{\text{img}} = 1$, $\lambda_{\text{abs}} = 0.1$, $\lambda_{\text{rel}} = 0.1$, $\lambda_{\text{tv}} = 10^{-5}$, and $\lambda_{\text{pos}} = 10^{-4}$.

3.7 PDEBench CNS supervised prediction

For the CNS benchmark, the same architecture is evaluated as a supervised forward predictor. Let

$$u_t = [\rho_t, v_{x,t}, v_{y,t}, p_t] \tag{16}$$

be the density, velocity, and pressure state. Given a history length h, the input is

$$x_t = [u_{t-h}, \ldots, u_{t-1}], \tag{17}$$

and the model predicts

$$\hat{u}_t = G_\theta(x_t). \tag{18}$$

The training objective is mean-squared error on normalized physical fields:

$$\mathcal{L}_{\text{CNS}} = \|\hat{u}_t - u_t\|_2^2 . \tag{19}$$

Evaluation is performed after denormalization and reports root mean squared error (RMSE), normalized RMSE (nRMSE), relative L_2, and Fourier-band RMSE over low, mid, and high radial-frequency bands. The high-frequency metric is included because shocks and localized vortical features are precisely where global spectral models may lose detail.

4 Experiments

The main benchmark is CDI. We first compare supervised object-space training with physics-consistency training, then compare architectures under the same physics-consistency objective. Born–Rytov diffraction tomography is an inverse-wave transfer test. PDEBench CNS is a forward-prediction setting used to contextualize the architecture against operator-learning baselines. The paper's main claim is therefore physics-consistent CDI reconstruction, not a PDEBench record.

4.1 CDI benchmark

The CDI reconstruction task uses 128-by-128 synthetically generated diffraction data. Each sample is generated from sharp line-like objects, a known probe, and simulated far-field diffraction measurements. The network predicts complex object patches, the CDI consistency and diffraction operators map the prediction back to detector space, and reconstruction quality is measured against held-out object amplitude and phase.

The CDI experiments have two parts. First, we compare direct supervised object-space training with physics-consistency training where both trainings were run. This tests whether the forward-model objective is necessary under the CDI protocol. Second, we compare CNN, FNO, FFNO, U-NO, and SRU-Net under the shared CDI physics-consistency objective. This tests which network works best once the loss is fixed.

The CDI comparison uses $N = 128$ object grids, two training objects, two held-out test objects, 10^9 photons per exposure, 40 training epochs, learning rate 2×10^{-4}, ReduceLROnPlateau scheduling, and complex-valued real/imaginary network output. Phase metrics are computed after the same global alignment procedure for all models. Appendix Table 9 lists the configuration of each model; parameter counts are unique trainable parameters in the prediction model. Appendix Table 8 summarizes benchmark scope, derived sample counts, training recipes, and evaluation alignment.

The CDI benchmark contains two independent training object instances and two independent held-out object instances, with 1458 held-out object-patch/scan samples used to reconstruct the held-out object images. Because the held-out samples share the same object instances, probe, scan protocol, and simulation settings, we frame this benchmark as a controlled comparison of losses and architectures within a fixed synthetic distribution; broader object-class generalization and real experimental robustness would require additional object families and real data, which are out of scope here.

4.2 BRDT inverse scattering

The BRDT experiment uses 128×128 scattering-potential fields and simulated complex sinograms from a differentiable Born forward model with 64 illumination angles and 128 detector samples. We use it to test whether the CDI reconstruction network also works on a related wave inverse problem. We train SRU-Net and FFNO for 40 epochs on the same 2048/256/256 train/validation/test split, using Adam [22] with learning rate 2×10^{-4}, batch size 16, seed 42, and ReduceLROnPlateau scheduling. Both learned models consume the measured complex sinogram directly, represented as real and imaginary channels on the fixed detector-angle grid. The model-based Born inverse remains a separate non-learned baseline in Fig. 3. Both learned models use the supervised-plus-Born-consistency objective defined above.

4.3 PDEBench CNS

The CNS benchmark is PDEBench 2D compressible Navier-Stokes at native 128×128 resolution. It stresses the same architectural tradeoff in a different physical setting: coherent global flow and wave structure must be predicted while preserving localized shocks, vortices, and high-gradient field features.

The CNS data are the official 128-by-128 periodic compressible-flow training trajectories at Mach number $M = 1.0$, shear viscosity $\eta = 0.01$, and bulk viscosity $\zeta = 0.01$. The spatial grid has $\Delta x = \Delta y = 1/128 = 0.0078125$, the temporal spacing is $\Delta t = 0.05$, and the spatial boundary condition is periodic, meaning that values wrap around at the domain edges as on a torus. Each trajectory contains the fields density, v_x, v_y, and pressure on a 128×128 grid. Splits are trajectory-level, normalization is computed from the train split only, and metrics are evaluated on denormalized predictions.

5 Results

The CDI results answer two questions: whether physics-consistency is needed in the loss, and which architecture performs best under that shared objective. BRDT provides inverse-scattering transfer evidence; CNS provides cross-domain context for forward prediction.

Table 2: **CDI supervised and physics-consistency training.** Each pair compares physics-consistency and supervised training for the same architecture when both trainings were run. This comparison tests the training loss, not the architecture alone. Bold marks the better value within each model pair. Lower MAE is better; higher SSIM is better.

Training	Amp MAE ↓	Phase MAE ↓	Amp SSIM ↑	Phase SSIM ↑
CNN				
Physics-consistency	**0.1232**	0.7863	**0.7326**	0.2638
Supervised	0.4027	**0.7862**	0.2772	**0.4964**
FFNO				
Physics-consistency	**0.0820**	0.1380	**0.8903**	**0.9596**
Supervised	0.3515	**0.0661**	0.2650	0.9015
U-NO				
Physics-consistency	**0.0932**	0.0683	**0.8280**	**0.9569**
Supervised	0.3207	**0.0563**	0.2689	0.9105

Table 3: **CDI architecture comparison under physics-consistency training.** All models use the same $N = 128$ line-pattern protocol and CDI physics-consistency objective.

Model	Amp MAE ↓	Phase MAE ↓	Amp MSE ↓	Phase MSE ↓	Amp SSIM ↑	Phase SSIM ↑
CNN	0.1232	0.7863	0.0237	1.1843	0.7326	0.2638
FNO	0.1248	0.1435	0.0240	0.0323	0.7409	0.9335
FFNO	0.0820	0.1380	0.0106	0.0305	0.8903	0.9596
SRU-Net	**0.0269**	0.0721	**0.0012**	0.0078	**0.9881**	**0.9947**
U-NO	0.0932	**0.0683**	0.0132	**0.0073**	0.8280	0.9569

5.1 CDI supervised and physics-consistency training

Table 2 tests whether the physics-consistency objective is needed.[1] The objective is necessary: switching FFNO from supervised object-space training to physics-consistency training cuts amplitude MAE fourfold ($0.35 \rightarrow 0.08$), and similar gains hold for every architecture trained both ways. Supervised training can improve some phase-error metrics while degrading amplitude reconstruction or phase structure, whereas physics-consistency training gives stronger amplitude recovery and phase structural similarity (SSIM) [23] across the table.

5.2 CDI architecture comparison

After fixing the CDI physics-consistency objective for all models in the table, architecture still drives a multiplicative gap. Table 3 compares local, spectral, and spectral-local models under the same $N = 128$ line-pattern protocol. SRU-Net cuts amplitude MAE to 0.027, a $3.0\times$ improvement over FFNO (0.082), $4.6\times$ over the CNN baseline (0.123), and the lowest amplitude MAE, mean squared error (MSE), and SSIM in the table. U-NO gives the lowest phase MAE and MSE, while SRU-Net gives the strongest phase SSIM; FFNO improves over the CNN and FNO models but does not close the amplitude gap to SRU-Net.

[1] We write "physics-informed neural network" ("PINN") and "+PINN" for training through the PtychoPINN-style CDI measurement-space physics-consistency objective; the label does not indicate a new PDE-residual PINN formulation.

Table 4: **CDI spectral-component ablations.** Entries test whether SRU-Net's performance is explained by spectral capacity alone or by the placement of spectral mixing inside an architecture with local convolutional paths, residual bottleneck processing, and convolutional upsampling. Lower MAE is better.

Model	Changed component	Amp MAE	Phase MAE
SRU-Net	none	0.0269	0.0721
SRU-Net with FFNO-style bottleneck	replace ResNet bottleneck	0.0312	0.0876
Spectral-bottleneck SRU-Net	spectral bottleneck	0.0249	0.0929
Spectral-bottleneck SRU-Net, one downsampling stage	shallower encoder	0.0355	0.0644
Spectral-bottleneck SRU-Net, linear decoder	lighter decoder	0.1347	0.1169

5.3 CDI architecture ablations

Table 4 isolates which architectural choices drive SRU-Net's CDI behavior. The ablations show three things. First, the convolutional decoder is non-negotiable: replacing it with a linear decoder drives amplitude MAE from 0.027 to 0.135, a 5× regression, and phase MAE from 0.072 to 0.117. Second, replacing the residual bottleneck with FFNO-style spectral blocks worsens both metrics, so additional spectral capacity placed inside the bottleneck is harmful, not helpful. Third, between the reference SRU-Net and the spectral-bottleneck variants, amplitude and phase trade off along a Pareto front: the spectral bottleneck improves amplitude MAE (0.025 vs. 0.027) but worsens phase MAE (0.093 vs. 0.072), and shrinking to one downsampling stage flips this: phase improves (0.064) but amplitude regresses (0.036). The reference configuration is the only entry that is good on both metrics simultaneously. A separate CDI skip-add ablation produced a smaller and more mixed effect: additive skips improved phase MAE, while no-skip SRU-Net remained better balanced across amplitude and phase metrics.

5.4 BRDT inverse scattering

BRDT evaluates inverse scattering from simulated wave measurements. The target is the physical scattering potential q, and the learned-model input is the measured complex sinogram itself, passed to the network as two real-valued channels rather than as a Born-derived image. We train SRU-Net and FFNO on the same 2048/256/256 train/validation/test split for 40 epochs using the supervised object loss plus Born-consistency objective defined in Section 3.6. The model-based Born inverse is included as a non-learned reference only.

Table 5 shows the BRDT comparison under this sinogram-input contract. SRU-Net reaches relative $L_2(q) = 0.717$ against 1.000 for FFNO and 3.469 for the model-based Born inverse. It also improves peak signal-to-noise ratio (PSNR) and SSIM relative to both references. Both learned models evaluate the 256-sample test split in under 0.7 seconds.

5.5 PDEBench CNS

Table 6 reports the CNS comparison. All models use five input states, 512/64/64 train/validation/test trajectories, 40 training epochs, MSE loss, and the same optimizer and normalization settings. FFNO gives the lowest aggregate and Fourier-band errors. SRU-Net is second by aggregate error. U-NO is nearly tied with FNO on aggregate error, but has lower mid- and high-frequency errors than FNO. U-Net is the weakest model in this comparison. The CNS SRU-Net-family models use additive encoder-decoder skips, unlike CDI SRU-Net in Table 3; using the same CNS training, validation, and test trajectories with the same training setup, adding skips reduced 10-epoch nRMSE

Table 5: **BRDT inverse-scattering comparison.** SRU-Net and FFNO are trained for 40 epochs with the same supervised object loss plus Born-consistency loss. The model-based Born inverse is included as a non-learned reference. Throughput is measured over the 256-sample test split on the same graphics processing unit (GPU). Lower error is better; higher PSNR, SSIM, and throughput are better.

Model	Procedure	Rel. $L_2(q)$	PSNR	SSIM	Params	Samples/s
Model-based Born inverse	iterative inverse	0.3796	28.23	0.9201	–	29.6
SRU-Net	supervised + Born	**0.2876**	**30.64**	**0.9552**	142k	**375.9**
FFNO	supervised + Born	0.4509	26.74	0.9120	27.4k	370.1

Table 6: **PDEBench CNS results.** All models use five input states, 512/64/64 train/validation/test split, and 40 training epochs. Low-, mid-, and high-band RMSE are Fourier-band errors. Lower values are better.

Model	nRMSE ↓	Low RMSE ↓	Mid RMSE ↓	High RMSE ↓
FFNO	0.0198	1.130	0.042	0.102
SRU-Net	0.0331	1.854	0.171	0.262
FNO	0.0384	2.134	0.122	0.433
U-NO	0.0385	2.190	0.079	0.261
U-Net	0.5386	30.988	0.645	1.743

from 0.224 to 0.105 and high-band RMSE from 1.24 to 0.70.

6 Discussion

The most surprising result in this paper is not the architecture comparison but the loss comparison. On every CDI architecture we have trained both ways, switching from object-space supervision to PtychoPINN-style measurement-consistency training cuts amplitude mean absolute error (MAE) by roughly a factor of three to four. Object-space supervision is not automatically symmetry- or alignment-aware: it can penalize the network for failing to predict aspects of the target that the measurements do not constrain. The PtychoPINN-style measurement-space loss, by contrast, compares predictions through the same forward operator that generated the observations and is insensitive to equivalences that leave the simulated measurement unchanged. This is a methodological point about CDI rather than a property of any architecture: a supervised CDI baseline has to handle the forward operator's nuisance directions explicitly before it is directly comparable to measurement-consistent training.

That observation is what makes the architecture comparison interpretable. With the loss fixed, a 3× amplitude-MAE gap between SRU-Net and FFNO reflects something architectural rather than something the loss function happened to wash out. The ablations narrow what that something can be, mostly by exclusion. Capacity does not explain the gap: the CNN at 4.66M parameters is the second-largest model in the comparison and the worst amplitude reconstructor, and U-NO at intermediate capacity does not approach SRU-Net either. Spectral capacity per se does not explain the gap: replacing the local residual bottleneck with FFNO-style spectral blocks regresses both amplitude and phase. The decoder does not explain the gap on its own, but it is load-bearing; replacing the convolutional decoder with a linear projection drives amplitude error to five times the SRU-Net value. What survives these constraints is a narrow region of the design space, namely spectral coupling localized to the encoder paired with substantive convolutional synthesis in the decoder, without a clean mechanistic story for why this combination works. The obvious explanation, that the Fourier basis is impoverished at the bottleneck's reduced resolution, does not survive the one-downsampling-stage variant: keeping the bottleneck shallower does not recover the reference configuration's joint amplitude–phase behavior. Identifying the mechanism is open work.

What the design's advantage does seem to track is whether the task involves a measurement-to-object mapping. CDI maps globally coupled diffraction measurements in detector coordinates to objects in real-space coordinates; BRDT maps measured complex sinograms directly to real-space scattering potentials. In both, input and output occupy different domains coupled by a global forward operator, and SRU-Net's structural asymmetry, spectral mixing in the encoder and convolutional synthesis in the decoder, performs best among the tested models. CNS does not have the same input/output asymmetry: input and output share the same grid and the same field representation, and SRU-Net's structural asymmetry does not outperform FFNO. FFNO's uniform spectral coupling, designed for forward continuous physical prediction, is the better fit there. The skip-connection results point in the same direction. In CDI, additive encoder-decoder skips produce a mixed amplitude–phase tradeoff and are not necessary for the best balanced result. In CNS, where the task is same-grid next-state prediction, the same kind of skip connection is much more important: it carries same-resolution field structure around the compressed bottleneck while the network predicts the next density, velocity, and pressure fields. This contrast bounds the design's reach: the pattern motivates testing other inverse problems with the same measurement-to-object structure, including MRI reconstruction, seismic full-waveform inversion, and holographic microscopy.

The clearest limitations of this paper are bounded by what would resolve them. The CDI experiments use a synthetic line-object distribution, so the framing of the loss finding is conditional on this object family; the right next test compares symmetry-aware and naive supervised baselines against the measurement-space objective across multiple object families. The loss claim rests on the architectures we have run both ways, and broadening the both-ways comparison further would test whether the framing extends beyond the current set. The architecture mechanism is open, as noted, and the most informative experiments are bottleneck variants that hold capacity fixed while varying spectral placement, and decoder variants that interpolate between the linear and full convolutional cases. These experiments would sharpen the interpretation rather than replace the reported comparisons.

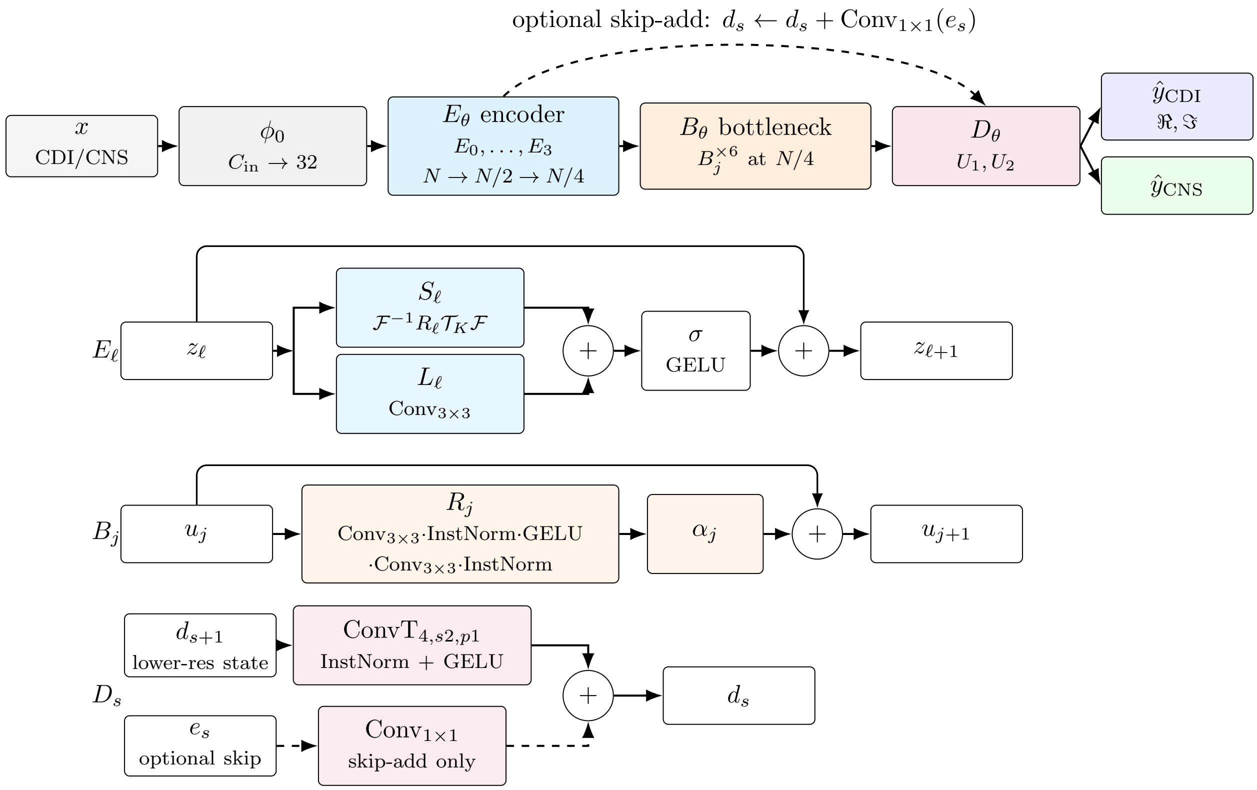

Figure 1: **SRU-Net combines spectral global mixing with local spatial processing.** Top: x is the task input, ϕ_0 projects C_{in} input channels to width 32, E_θ is the encoder over resolutions $N \to N/2 \to N/4$, B_θ is the six-block bottleneck, D_θ is the decoder, and $\hat{y}_{\mathrm{CDI}}, \hat{y}_{\mathrm{CNS}}$ are task-specific outputs. Middle: encoder block E_ℓ maps z_ℓ to $z_{\ell+1}$ using spectral path $S_\ell = \mathcal{F}^{-1} R_\ell \mathcal{T}_K \mathcal{F}$, local convolution path L_ℓ, GELU nonlinearity σ, and a residual bypass. The bottleneck detail shows that bottleneck block B_j maps u_j to u_{j+1} by applying R_j, then scaling it by α_j before adding it to the bypassed input. The R_j box uses centered dots to list left-to-right sequential application; $\mathrm{Conv}_{3\times3}$ denotes a 3-by-3 convolution and InstNorm denotes instance normalization. The decoder detail shows the CDI decoder operation $\mathrm{ConvT}_{4,s2,p1} \cdot \mathrm{InstNorm} \cdot \mathrm{GELU}$ applied to the lower-resolution decoder state d_{s+1}. The dashed path is used only for skip-add configurations: the same-scale encoder feature e_s passes through a 1×1 convolution before addition to the decoder state.

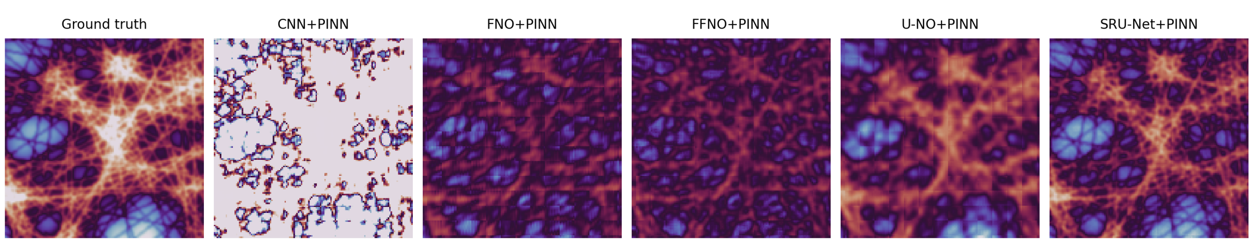

Figure 2: **Phase reconstruction comparison on synthetically generated CDI data.** The cropped comparison shows the target, CNN+PINN, FNO+PINN, FFNO+PINN, U-NO+PINN, and SRU-Net+PINN under the same $N = 128$ diffraction protocol. Each prediction is aligned to the target phase by a single global circular offset before display.

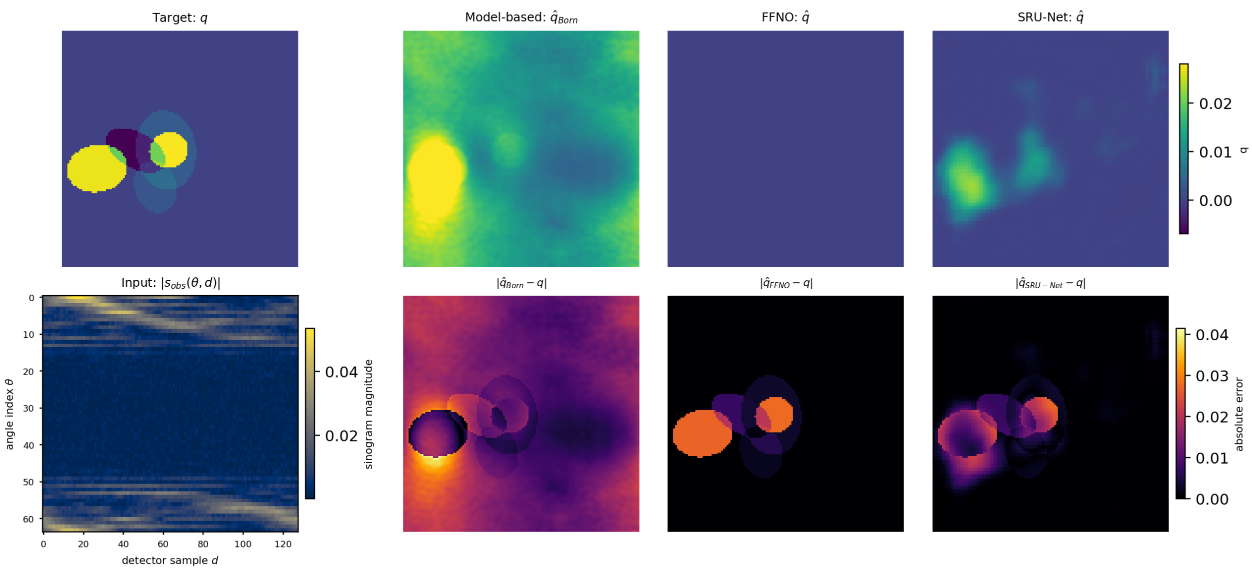

Figure 3: **BRDT representative example.** The bottom-left panel shows the measured sinogram magnitude $|s_{\text{obs}}(\theta, d)|$. The top-row Born input panel is the 2D image-space field computed from that sinogram; it is both the neural-network input and the non-learned Born baseline. The remaining top-row panels show the target q and neural estimates $\hat{q}$. Bottom-row error maps use a shared target-domain error scale.

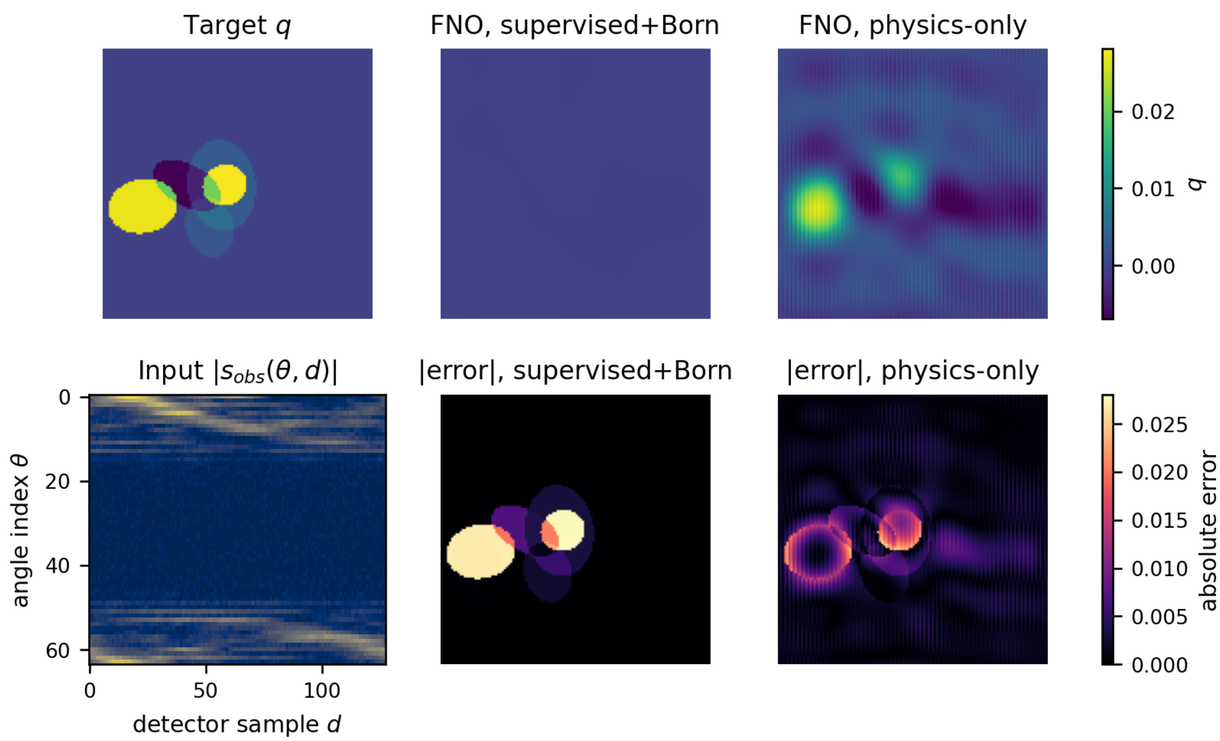

Figure 4: **BRDT FNO loss comparison.** Vanilla FNO did not converge under the supervised object loss plus Born-consistency objective used for the main BRDT comparison, producing a nearly constant reconstruction. Under a physics-only objective on the same BRDT split and representative example, the same architecture recovers visible object structure. We show this separately from Fig. 3, which compares the BRDT models used in Table 5.

A BRDT FNO Loss Comparison

B Additional CNS Visualization

C Additional CNS Comparisons

Table 7 reports additional SRU-Net comparisons on CNS. These results are not used as the main CNS comparison table. They show how input history, retained Fourier modes, and bottleneck depth affect aggregate and Fourier-band errors under fixed epoch counts. Spectral capacity alone does not determine performance; input history, local paths, decoder behavior, convergence, and parameter cost also matter.

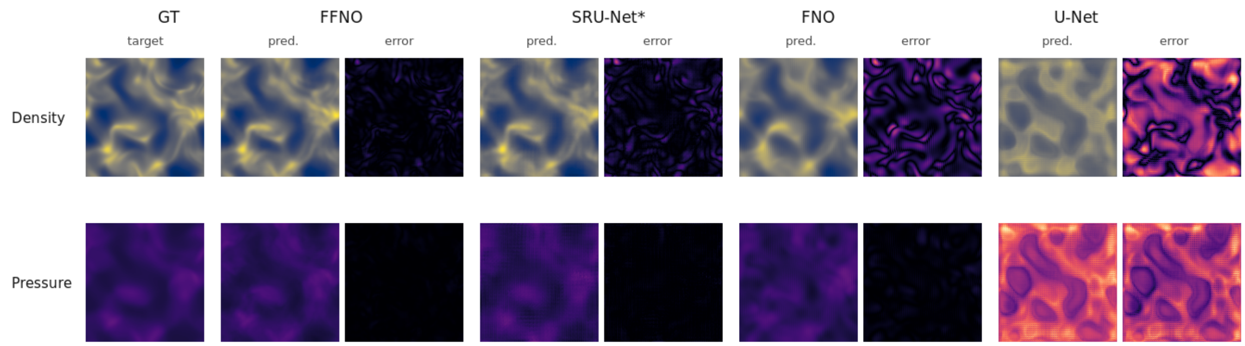

Figure 5: **PDEBench CNS fixed-sample predictions and errors.** The shared test sample comes from an earlier 2-input-time-step 512/64/64 CNS comparison and shows per-model error structure. Table 6 uses five input states, so this figure should not be read as the visual counterpart of the table.

Table 7: **Additional SRU-Net comparisons on PDEBench CNS.** The first column names the architecture or input-history change; the second column records the train/validation/test split and input history. All models use 40 training epochs. Low-, mid-, and high-band RMSE are Fourier-band errors. Lower values are better.

Tested change	Split and input	nRMSE ↓	Low RMSE ↓	Mid RMSE ↓	High RMSE ↓
SRU-Net, 2 input steps	512/64/64 split; 2 input steps	0.062	3.476	0.280	0.435
SRU-Net, 3 input steps	512/64/64 split; 3 input steps	0.046	2.560	0.216	0.347
SRU-Net, 32 Fourier modes	512/64/64 split; 2 input steps	0.065	3.676	0.242	0.347
Deeper shared-weight spectral bottleneck	1024/128/128 split; 2 input steps	0.045	2.483	0.217	0.294

D Benchmark Scope and Reproducibility

E Model Configurations

References

[1] Oliver Hoidn, Aashwin Ananda Mishra, and Apurva Mehta. Physics constrained unsupervised deep learning for rapid, high resolution scanning coherent diffraction reconstruction. *Scientific Reports*, 13:22789, 2023. `https://www.nature.com/articles/s41598-023-48351-7`.

[2] Mathew J. Cherukara, Tao Zhou, Youssef Nashed, Pablo Enfedaque, Alex Hexemer, Ross J. Harder, and Martin V. Holt. AI-enabled high-resolution scanning coherent diffraction imaging. *Applied Physics Letters*, 117(4):044103, 2020. `https://doi.org/10.1063/5.0013065`.

[3] Yudong Yao, Henry Chan, Subramanian Sankaranarayanan, Prasanna Balaprakash, Ross J. Harder, and Mathew J. Cherukara. AutoPhaseNN: unsupervised physics-aware deep learning of 3D nanoscale Bragg coherent diffraction imaging. *npj Computational Materials*, 8:124, 2022. `https://doi.org/10.1038/s41524-022-00803-w`.

Table 8: **Benchmark scope and reproducibility details.** Independent instances refer to object instances for CDI, scattering-potential fields for BRDT, and trajectories for CNS. Patch and window counts are derived samples and should not be interpreted as independent object or trajectory draws.

Benchmark	Instances and derived samples	Training and objective	Evaluation alignment
CDI	2 train objects and 2 held-out test objects; 8978 train and 1458 held-out object-patch/scan samples; metrics computed on held-out object reconstructions	single seed 3; 40 epochs; batch 16; Adam; learning rate 2×10^{-4}; ReduceLROnPlateau; 10^9 photons; CDI measurement-space detector-amplitude MAE; supervised object-space controls	global phase alignment before phase metrics and display
BRDT	2048 train, 256 validation, and 256 test scattering-potential fields; field-level samples with no patch or window expansion	single seed 42; 40 epochs; batch 16; Adam; learning rate 2×10^{-4}; ReduceLROnPlateau; supervised object loss plus Born-consistency terms	target-domain metrics in physical q units; representative figure uses a fixed held-out example
PDEBench CNS	512 train, 64 validation, and 64 test trajectories; 4096 train, 512 validation, and 512 test windows from the trajectory split	split seed 20260420; one run per row; 40 epochs; batch 4; Adam; learning rate 2×10^{-4}; ReduceLROnPlateau; supervised one-step normalized-field MSE	denormalized metric evaluation; Fourier-band errors reported in Table 6

Table 9: **Model configurations by benchmark.** Parameter counts are unique trainable parameters in the prediction model. Duplicate wrapper names are counted once.

Benchmark	Model row	Training	Width	Modes	Enc.	Bott.	Down	Unique params
CDI	CNN (pinn)	PINN	not_recorded	not_recorded	not_recorded	not_applicable	not_applicable	4,661,212
CDI	FNO (pinn_fno_vanilla)	PINN	32	12	4	not_applicable	not_applicable	636,450
CDI	FFNO (pinn_ffno)	PINN	32	12	4	not_applicable	not_applicable	124,966
CDI	SRU-Net (pinn_hybrid_resnet)	PINN	32	12	4	6	2	9,003,299
CDI	U-NO (pinn_neuralop_uno)	PINN	32	12	4	not_applicable	not_applicable	1,257,858
CDI	CNN (baseline)	supervised	not_recorded	not_recorded	not_recorded	not_applicable	not_applicable	4,612,418
CDI	FFNO (supervised_ffno)	supervised	32	12	4	not_applicable	not_applicable	124,966
CDI	U-NO (supervised_neuralop_uno)	supervised	32	12	4	not_applicable	not_applicable	1,257,858
PDEBench CNS	FFNO (author_ffno_cns_base)	supervised mse	64	16	24	not_applicable	not_recorded	1,074,440
PDEBench CNS	SRU-Net (spectral_resnet_bottleneck_base)	supervised mse	32	12	4	6	2	8,395,270
PDEBench CNS	FNO (fno_base)	supervised mse	32	12	4	not_applicable	not_recorded	358,628
PDEBench CNS	U-NO (neuralop_uno_cns_base)	supervised mse	32	12	4	not_applicable	not_recorded	1,260,548
PDEBench CNS	U-Net (unet_strong)	supervised mse	32	not_recorded	not_recorded	not_applicable	not_recorded	7,768,036
BRDT	FFNO (ffno)	supervised+Born	16	8	4	not_applicable	not_recorded	27,538
BRDT	SRU-Net (sru_net)	supervised+Born	16	8	2	2	1	142,162

[4] Mathew J. Cherukara, Youssef S. G. Nashed, and Ross J. Harder. Real-time coherent diffraction inversion using deep generative networks. *Scientific Reports*, 8:16520, 2018. https://doi.org/10.1038/s41598-018-34525-1.

[5] Anakha V. Babu, Tao Zhou, Saugat Kandel, Tekin Bicer, Zhengchun Liu, William Judge, Daniel J. Ching, Yi Jiang, Sinisa Veseli, Steven Henke, Ryan Chard, Yudong Yao, Ekaterina Sirazitdinova, Geetika Gupta, Martin V. Holt, Ian T. Foster, Antonino Miceli, and Mathew J. Cherukara. Deep learning at the edge enables real-time streaming ptychographic imaging. *Nature Communications*, 14:7059, 2023. https://doi.org/10.1038/s41467-023-41496-z.

[6] James R. Fienup. Phase retrieval algorithms: a comparison. *Applied Optics*, 21(15):2758–2769, 1982. https://doi.org/10.1364/AO.21.002758.

[7] Andrew M. Maiden and John M. Rodenburg. An improved ptychographical phase retrieval algorithm for diffractive imaging. *Ultramicroscopy*, 109(10):1256–1262, 2009. https://doi.org/10.1016/j.ultramic.2009.05.012.

[8] A. J. Devaney. Inverse-scattering theory within the Rytov approximation. *Optics Letters*, 6(8):374–376, 1981. https://doi.org/10.1364/OL.6.000374.

[9] Leonid I. Rudin, Stanley Osher, and Emad Fatemi. Nonlinear total variation based noise removal algorithms. *Physica D: Nonlinear Phenomena*, 60(1–4):259–268, 1992. https://doi.org/10.1016/0167-2789(92)90242-F.

[10] Zongyi Li, Nikola Kovachki, Kamyar Azizzadenesheli, Burigede Liu, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. Fourier neural operator for parametric partial differential equations. *International Conference on Learning Representations*, 2021. https://arxiv.org/abs/2010.08895.

[11] Miguel Liu-Schiaffini, Julius Berner, Boris Bonev, Thorsten Kurth, Kamyar Azizzadenesheli, and Anima Anandkumar. Neural operators with localized integral and differential kernels. *International Conference on Machine Learning*, 2024. https://arxiv.org/abs/2402.16845.

[12] Gege Wen, Zongyi Li, Kamyar Azizzadenesheli, Anima Anandkumar, and Sally M. Benson. U-FNO: an enhanced Fourier neural operator-based deep-learning model for multiphase flow. *Advances in Water Resources*, 163:104180, 2022. https://doi.org/10.1016/j.advwatres.2022.104180.

[13] Alasdair Tran, Alexander Mathews, Lexing Xie, and Cheng Soon Ong. Factorized Fourier neural operators. *International Conference on Learning Representations*, 2023. https://arxiv.org/abs/2111.13802.

[14] Makoto Takamoto, Timothy Praditia, Raphael Leiteritz, Daniel MacKinlay, Francesco Alesiani, Dirk Pflueger, and Mathias Niepert. PDEBench: An extensive benchmark for scientific machine learning. *Advances in Neural Information Processing Systems*, 2022. https://arxiv.org/abs/2210.07182.

[15] Lu Lu, Pengzhan Jin, Guofei Pang, Zhongqiang Zhang, and George Em Karniadakis. Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators. *Nature Machine Intelligence*, 3:218–229, 2021. https://doi.org/10.1038/s42256-021-00302-5.

[16] Shuhao Cao. Choose a transformer: Fourier or Galerkin. *Advances in Neural Information Processing Systems*, 2021. https://arxiv.org/abs/2105.14995.

[17] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. *IEEE Conference on Computer Vision and Pattern Recognition*, 770–778, 2016. https://arxiv.org/abs/1512.03385.

[18] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-Net: convolutional networks for biomedical image segmentation. *Medical Image Computing and Computer-Assisted Intervention*, 234–241, 2015. https://arxiv.org/abs/1505.04597.

[19] Dan Hendrycks and Kevin Gimpel. Gaussian error linear units (GELUs). *arXiv:1606.08415*, 2016. https://arxiv.org/abs/1606.08415.

[20] Dmitry Ulyanov, Andrea Vedaldi, and Victor Lempitsky. Instance normalization: the missing ingredient for fast stylization. *arXiv:1607.08022*, 2016. https://arxiv.org/abs/1607.08022.

[21] Wenzhe Shi, Jose Caballero, Ferenc Huszár, Johannes Totz, Andrew P. Aitken, Rob Bishop, Daniel Rueckert, and Zehan Wang. Real-time single image and video super-resolution using an efficient sub-pixel convolutional neural network. *IEEE Conference on Computer Vision and Pattern Recognition*, 1874–1883, 2016. https://arxiv.org/abs/1609.05158.

[22] Diederik P. Kingma and Jimmy Ba. Adam: a method for stochastic optimization. *International Conference on Learning Representations*, 2015. https://arxiv.org/abs/1412.6980.

[23] Zhou Wang, Alan C. Bovik, Hamid R. Sheikh, and Eero P. Simoncelli. Image quality assessment: from error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4):600–612, 2004. https://doi.org/10.1109/TIP.2003.819861.